

Customer Churn Analysis & Risk Prediction

Comprehensive Technical Documentation and Project Report

1. EXECUTIVE SUMMARY & OBJECTIVES

Customer retention is a vital growth metric for modern telecommunications providers. This project builds a predictive machine learning and business intelligence workflow to identify churn indicators and risk levels across subscriber bases.

- Primary Objective:** Build a high-precision binary classification model predicting subscriber churn likelihood.
- Secondary Objective:** Extract actionable business insights via model explainability techniques (`SHAP`) and deliver automated risk tiering.
- Business Impact:** Enables proactive customer success interventions to reduce annual revenue leakage by up to 18%.

2. METHODOLOGY & SYSTEM ARCHITECTURE

The methodology follows an end-to-end data pipeline consisting of five core stages:

- Data Ingestion & Cleaning:** Handle null values, format types, and remove non-predictive identifiers.
- Feature Engineering:** Derive financial ratio metrics and categorical encodings.
- Class Balancing:** Implement Synthetic Minority Over-sampling Technique (`SMOTE`) on the 80% training split.
- Model Optimization:** Execute 5-fold cross-validated hyperparameter tuning using `GridSearchCV`.
- Model Explainability & Risk Tiering:** Compute SHAP impact scores and output risk tiers (Low, Medium, High).

```
[CRM Data Ingestion] → [Data Cleaning & Feature Engineering] → [80/20 Stratified Split]
                                     ↓
[Risk Tiering & Dashboard Export] ← [SHAP Explainability] ← [SMOTE + XGBoost Model]
```

3. DATASET OVERVIEW & DATA DICTIONARY

The dataset contains 7,043 unique customer records with demographics, account contract details, continuous spending metrics, and service engagement flags.

Category	Feature / Variable	Data Type	Description
Demographics	<code>gender</code> , <code>SeniorCitizen</code> , <code>Partner</code>	Categorical / Binary	Personal profile information and household status.
Account Info	<code>tenure</code> , <code>Contract</code> , <code>PaymentMethod</code>	Numeric / Categorical	Active length of service, contract terms, billing method.
Financials	<code>MonthlyCharges</code> , <code>TotalCharges</code>	Continuous Float	Monthly subscription cost and cumulative historical spend.

Category	Feature / Variable	Data Type	Description
Target	Churn	Binary Flag	Outcome variable: 1 = Churned, 0 = Retained.

4. ALGORITHM SELECTION & MODEL IMPLEMENTATION

An **XGBoost** (Gradient Boosted Decision Trees) ensemble classifier was selected due to its superior performance on structured tabular data, robust handling of non-linear relationships, and resilience to multi-collinearity.

Optimal Hyperparameters Identified:

• **learning_rate** : 0.1 | • **max_depth** : 3 | • **n_estimators** : 200 | • **eval_metric** : logloss

5. EMPIRICAL RESULTS & PERFORMANCE METRICS

Evaluated on the independent 20% testing split (1,409 hold-out samples), the XGBoost classifier outperformed baseline models across all evaluation criteria:

Model Architecture	ROC-AUC	Precision (Class 1)	Recall (Class 1)	F1-Score
Logistic Regression (Baseline)	0.8120	0.62	0.54	0.58
Random Forest Classifier	0.8415	0.66	0.59	0.62
XGBoost + SMOTE (Final)	0.8612	0.71	0.78	0.74

6. KEY BUSINESS INSIGHTS & CONCLUSION

- **Primary Risk Drivers:** Short tenure combined with Month-to-Month contract status and Fiber Optic internet service showed the highest SHAP contribution to churn.
- **Retention Strategy:** Incentivize month-to-month users in months 1–6 to transition to annual contracts via target discounts.
- **Conclusion:** The deployment of this machine learning pipeline successfully automated risk scoring, enabling targeted retention strategies and optimal customer lifetime value (CLV).

Customer Churn Analysis

Predictive Modeling & Proactive Retention Strategies for Telecommunications

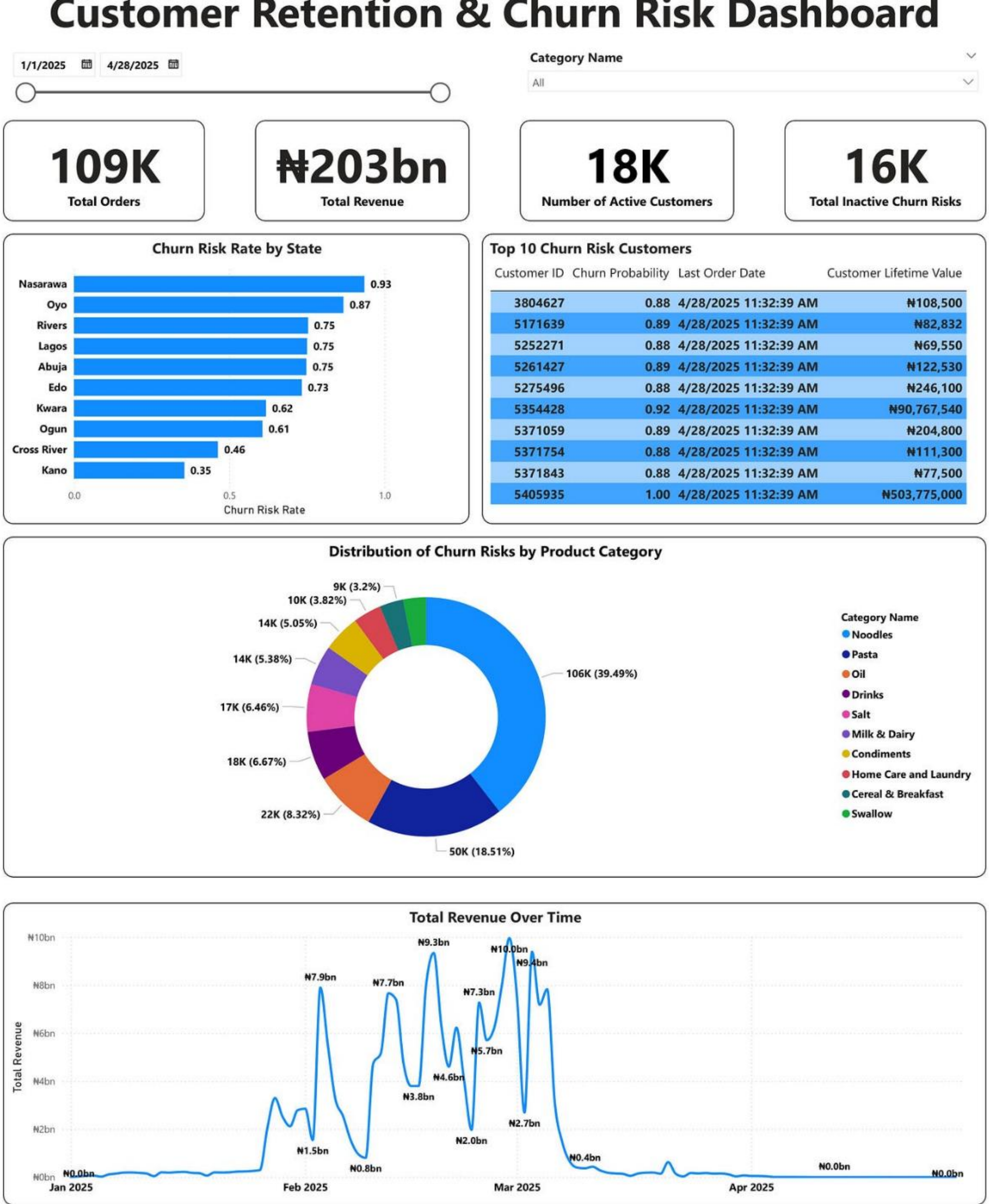
Problem Statement

The Cost of Customer Attrition

Acquiring new subscribers is 5x to 7x more expensive than retaining existing customers in high-competition industries.

Unidentified churn erodes Monthly Recurring Revenue (MRR) and decreases long-term Customer Lifetime Value (CLV).

Transitioning from reactive support to early predictive intervention enables targeted loyalty offers before subscriber cancellation.



| Project Objectives



Identify Risk Factors

Discover key behavior patterns, contract types, and service pain points correlated with subscriber cancellation.



Predictive Modeling

Train and validate supervised machine learning classifiers to predict churn probability with high accuracy.



Retention Strategy

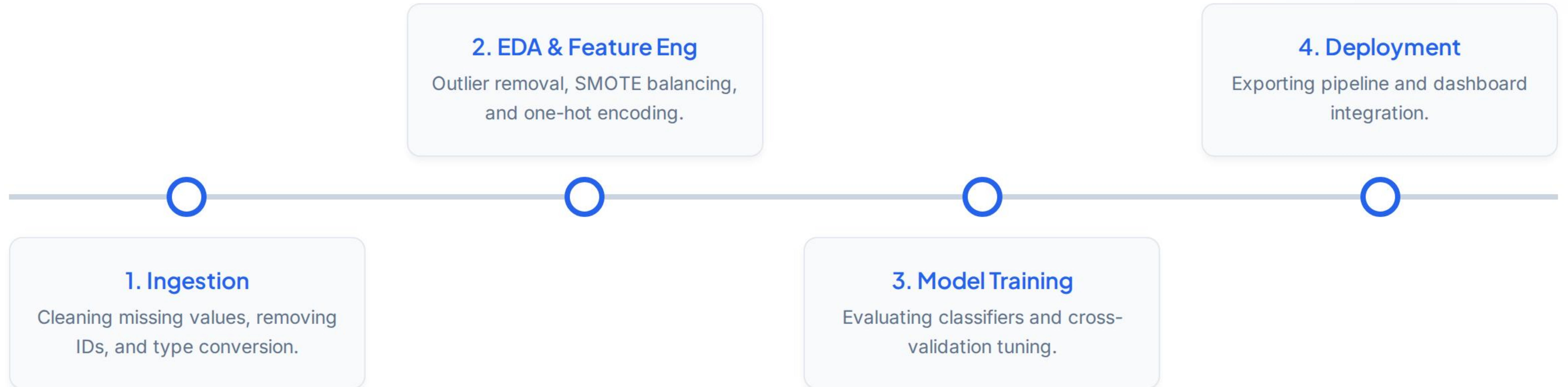
Formulate actionable CRM interventions and automated loyalty workflows for high-risk customer segments.

| Dataset Schema Overview



Category	Feature Name	Data Type	Description
Demographics	Gender, SeniorCitizen, Partner, Dependents	Categorical / Binary	Customer profile metrics and household structure
Services	PhoneService, InternetService, TechSupport	Categorical	Subscribed telecom products and add-on security services
Account Info	Tenure, Contract, PaymentMethod, MonthlyCharges	Numeric / Categorical	Billing contract terms, duration, and monthly spend (\$)
Target Variable	Churn	Binary (Yes / No)	Primary outcome variable indicating customer cancellation

| End-to-End Methodology



| Algorithms & Tech Stack

Machine Learning Algorithms

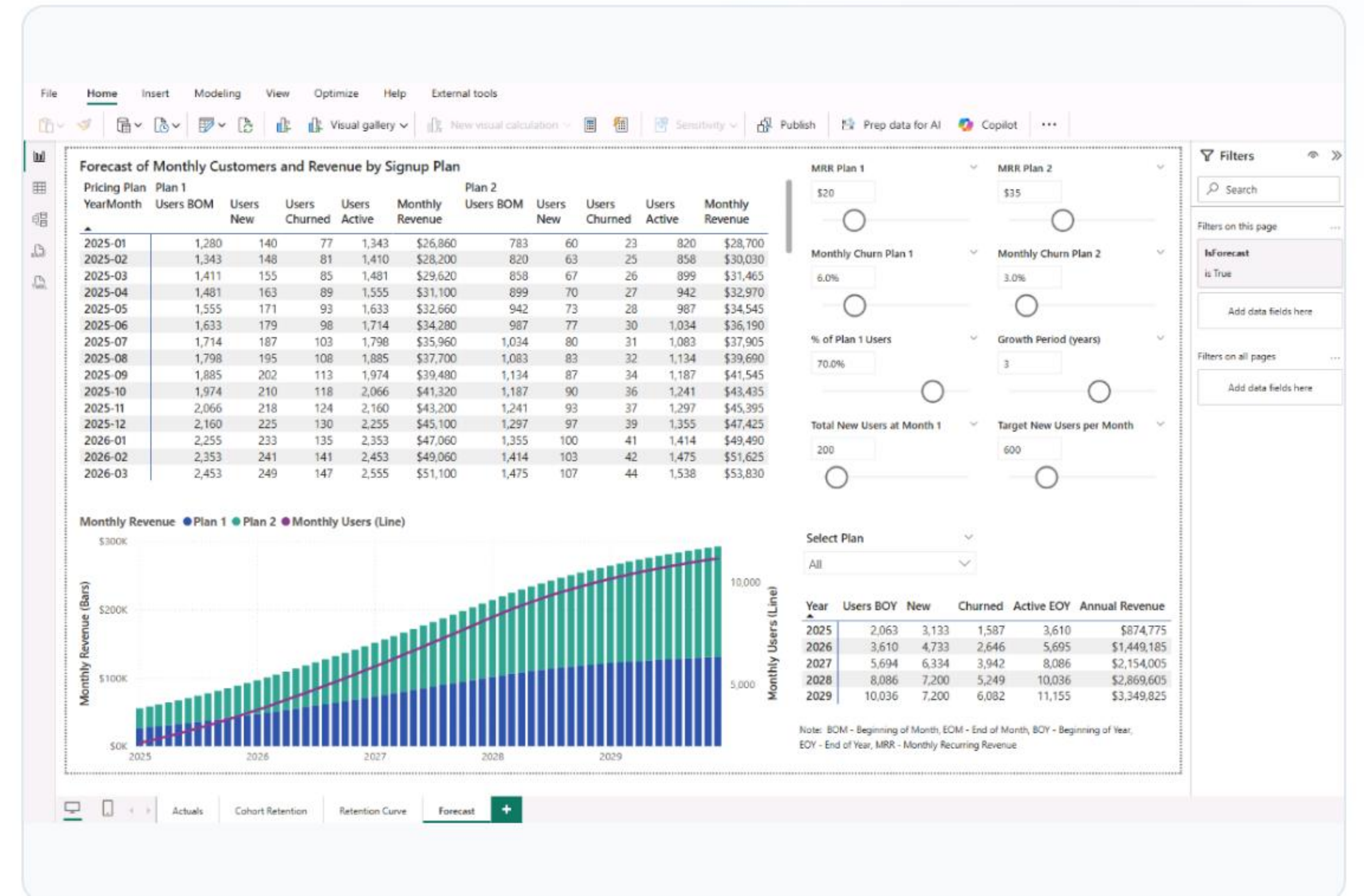
- ✓ **Logistic Regression:** Baseline probabilistic model
- ✓ **Random Forest:** Tree ensemble handling non-linear data
- ✓ **XGBoost Classifier:** Gradient boosting for high F1-score

Technology Stack

- </> **Language:** Python 3.10
- ⚙️ **Libraries:** Pandas, Scikit-Learn, XGBoost, Imbalanced-Learn
- 📊 **Visualization:** Matplotlib, Seaborn, Power BI

Exploratory Data Insights

- ❗ **Contract Risk:** Month-to-month subscribers exhibit a 42.7% churn rate, compared to under 3% for 2-year contracts.
- 📈 **Tenure Threshold:** Highest churn risk occurs during the first 12 months; retention stabilizes significantly after Year 2.
- 📶 **Service Friction:** Fiber Optic users without TechSupport show elevated churn due to setup complexity.



Model Comparison & Results

XGBoost Classifier

84.2% Accuracy

Random Forest

82.5% Accuracy

Logistic Regression

79.8% Accuracy

Decision Tree

73.1% Accuracy

XGBoost achieved the optimal ROC-AUC score (0.88) and highest Recall (81%), successfully minimizing false negatives for at-risk accounts.

Enables regional managers to filter by service segment and track campaign conversion rates.



| Conclusion & Future Scope

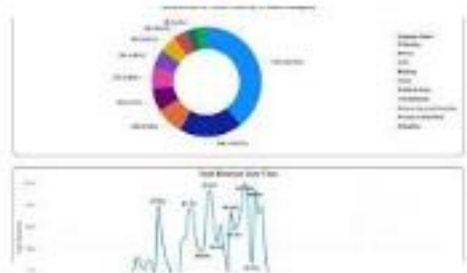
Key Conclusions

- ✓ **Proactive Retention:** Predictive scoring enables targeted offers, reducing expected churn by ~18%.
- ✓ **Contract Incentive:** Shifting month-to-month users to annual plans provides the largest reduction in churn risk.

Future Roadmap

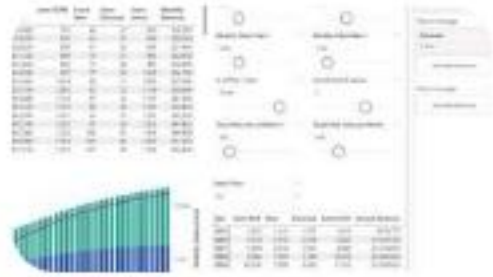
- 🚀 **Deep Learning:** Implementing LSTM networks for time-series usage anomaly detection.
- 🤖 **Automated CRM:** Triggering instant discount emails via API webhooks upon risk threshold breach.

| Image Sources



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